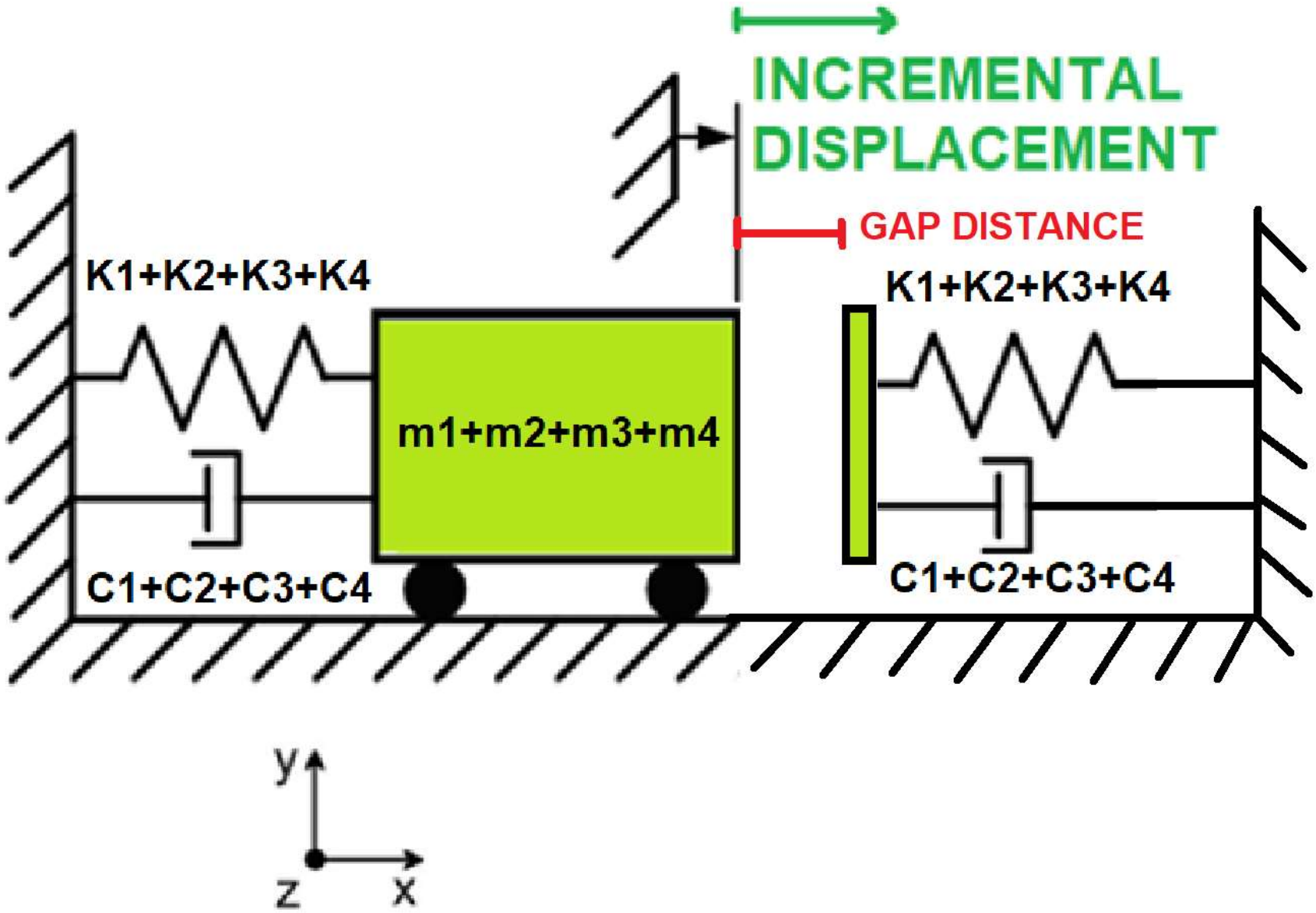
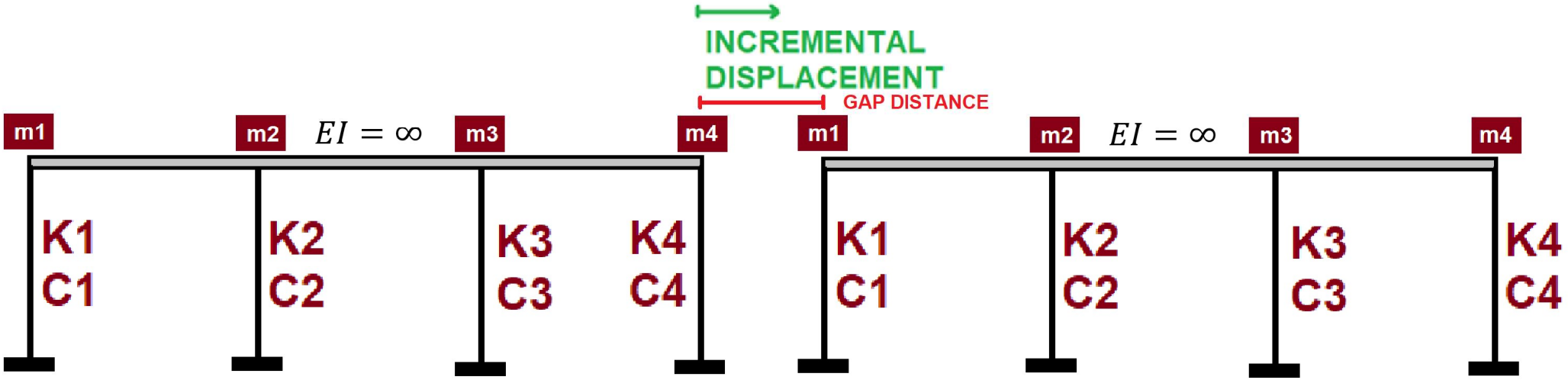
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

CONTACT-DRIVEN PUSHOVER ANALYSIS OF INELASTIC SDOF SYSTEMS: MONITORING PERIOD SHIFTS DURING SECONDARY SPRING ACTIVATION IN OPENSEES

WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)


INCREMENTAL
DISPLACEMENT
GAP DISTANCE
K1+K2+K3+K4
K1+K2+K3+K4
m1+m2+m3+m4
C1+C2+C3+C4
C1+C2+C3+C4
y
z
x


INCREMENTAL
DISPLACEMENT
GAP DISTANCE
m1
m2
$EI = \infty$
m3
m4
K1
C1
K2
C2
K3
C3
K4
C4
m1
m2
$EI = \infty$
m3
m4
K1
C1
K2
C2
K3
C3
K4
C4

Spyder (Python 3.12)

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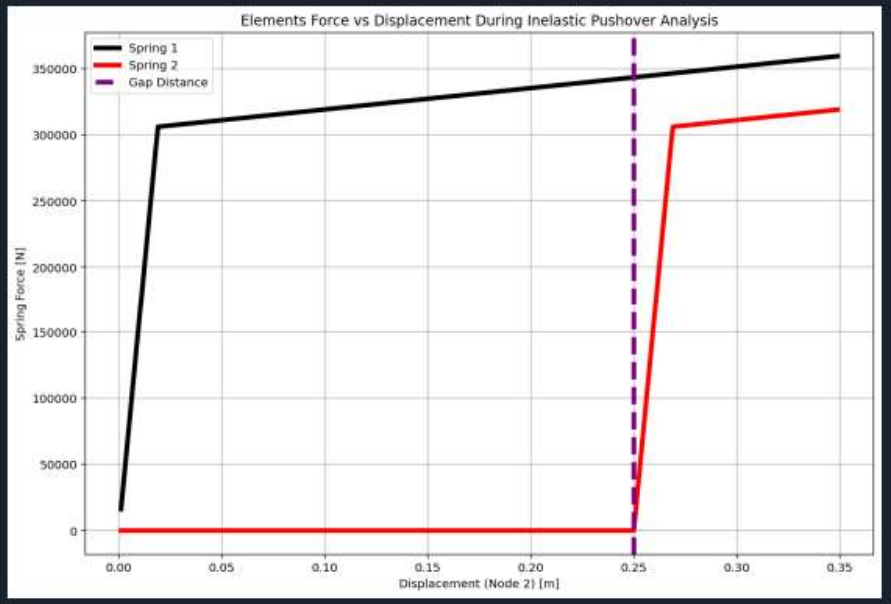
New file (Ctrl+N)

...SHOVER\CONTACT_PROBLEM_SDOF_PUSHOVER_MULTI_SPRING_PERIOD

C:\Users\Dell\Desktop\OPENSEES_FILES\+CONTACT_EXA...TACT_PROBLEM_SDOF_PUSHOVER_MULTI_SPRING_PERIOD.py

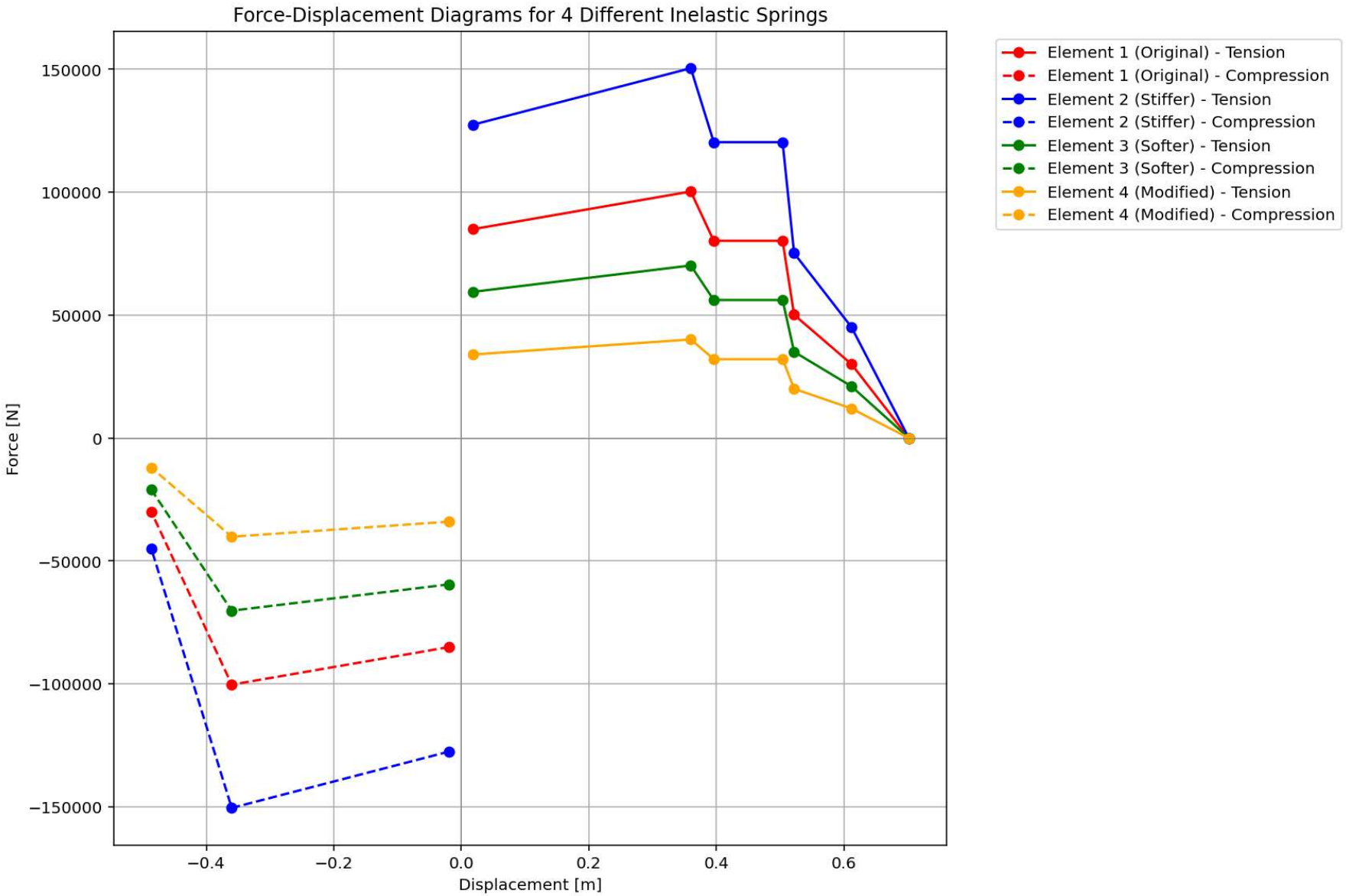
CONTACT_PROBLEM_SD...I_SPRING_PERIOD.py

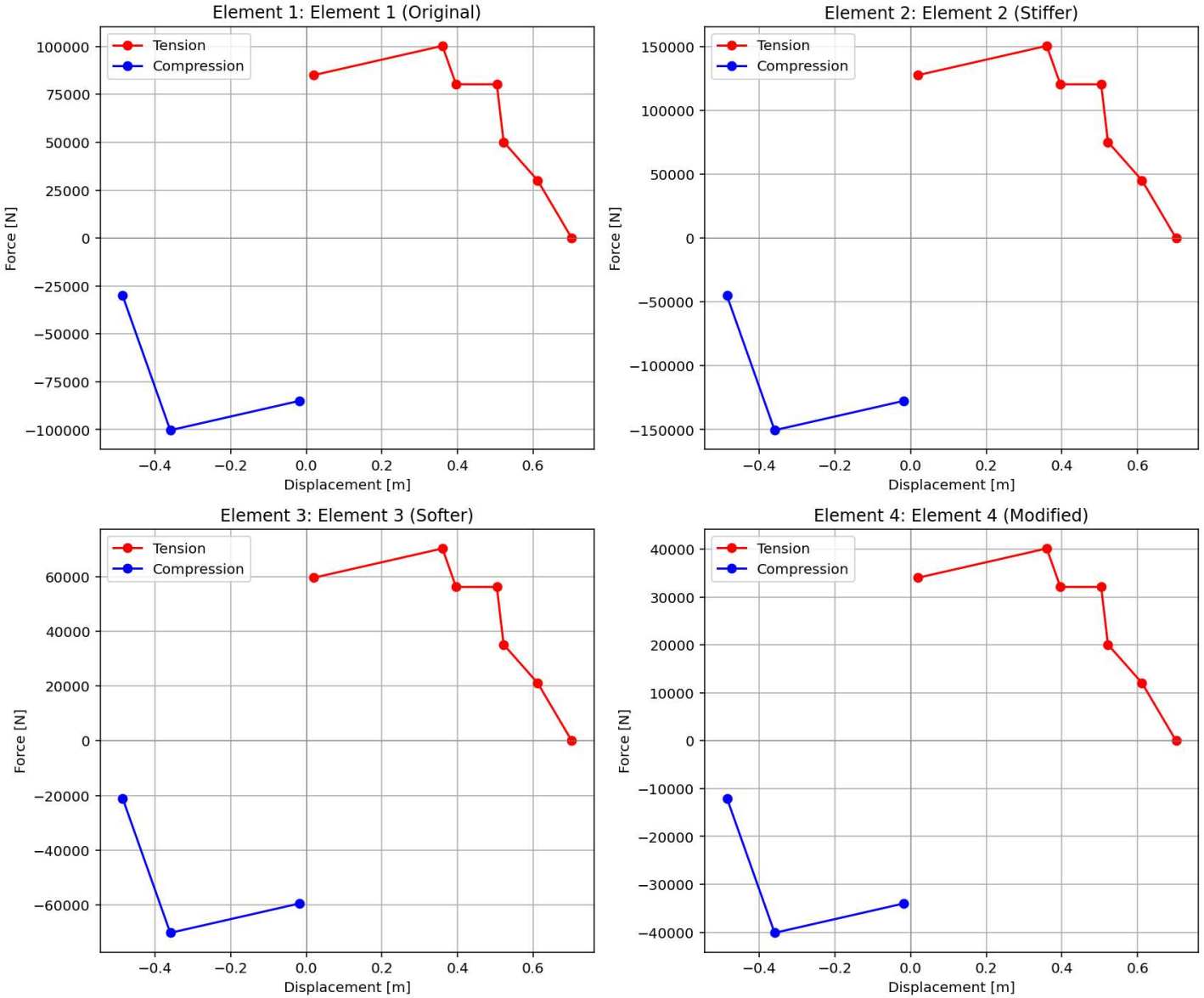
```
1  ##########################################################################
2  #                          >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE
3  #         CONTACT-DRIVEN PUSHOVER ANALYSIS OF INELASTIC SDOF SYSTEMS: MONIT
4  #                                        SECONDARY SPRING ACTIVATION IN OPE
5  #--------------------------------------------------------------------------
6  #                                THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHC
7  #                                       EMAIL: salar.d.ghashghaei@gmail.com
8  ##########################################################################
9  """
10 This script simulates the nonlinear pushover response of a single-degree-of
11  contact/gap mechanism. The structure has a primary spring (elastic or hyst
12  immediately, while a secondary parallel spring engages only when displacem
13  gap distance. This models structural components that come into contact onl
14  deformation thresholds, such as gap-opening in masonry infills, pounding b
15  or secondary bracing systems activating during strong seismic events.
16
17 The analysis tracks force-displacement response, stiffness degradation, and
18  as damage accumulates. The eigenvalue analysis at each step captures how t
19  increases with structural softening, a critical indicator of seismic vulne
20  progressive damage. Contact activation causes a sudden stiffness increase
21  followed by further period evolution as the system yields.
22 """
23 import openseespy.opensees as ops
24 import numpy as np
25 import matplotlib.pyplot as plt
26 import time as TI
27 import ANALYSIS_FUNCTION as S01
28 import PERIOD_FUN as S02
29 import EIGENVALUE_ANALYSIS_FUN as S04
30 import BILINEAR_CURVE as S05
31 #%%--------------------------------------------------------------------------
32 # Define parameters (units: m, N)
33 # Force-Displacement Relationship for Each Column
```

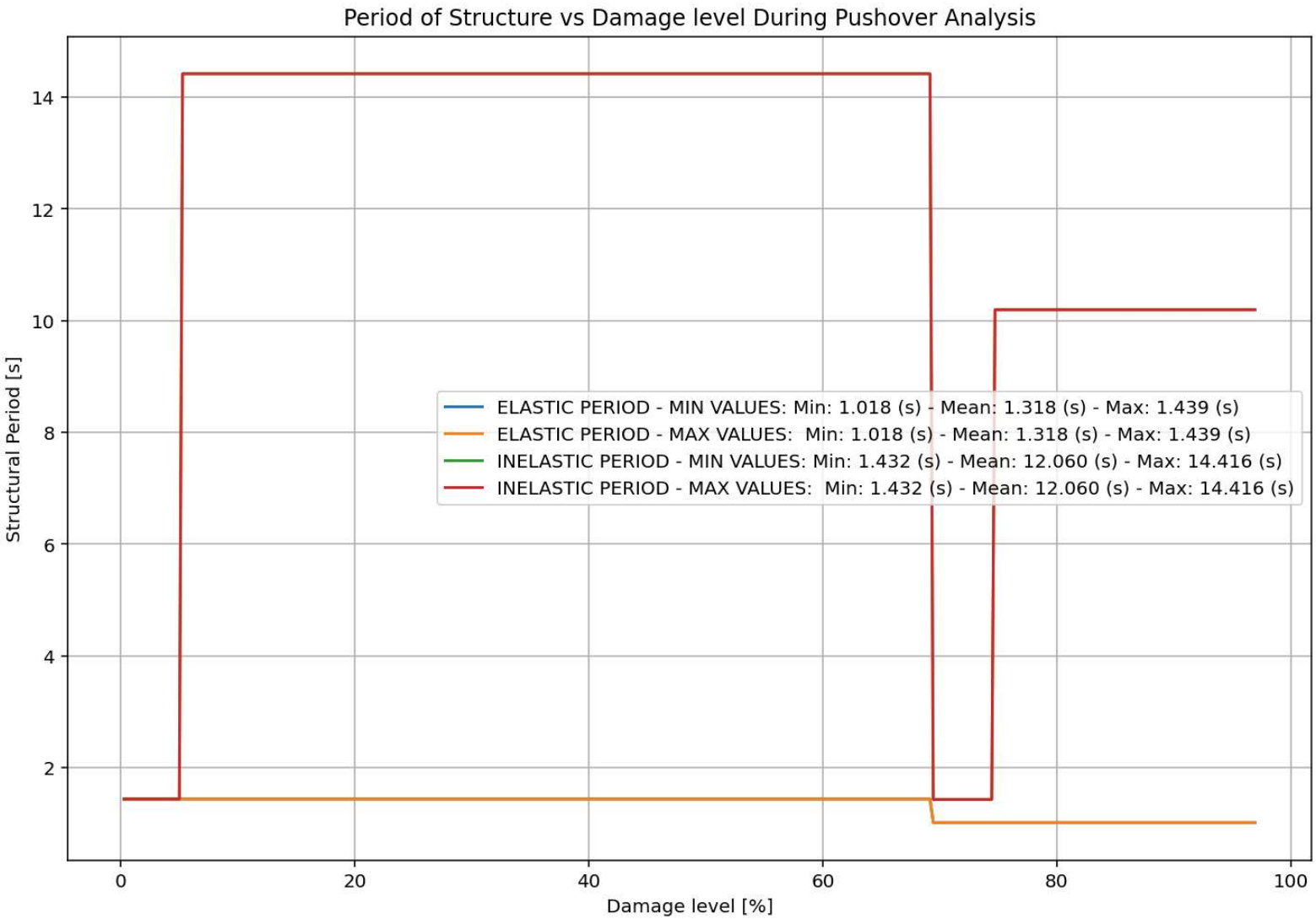


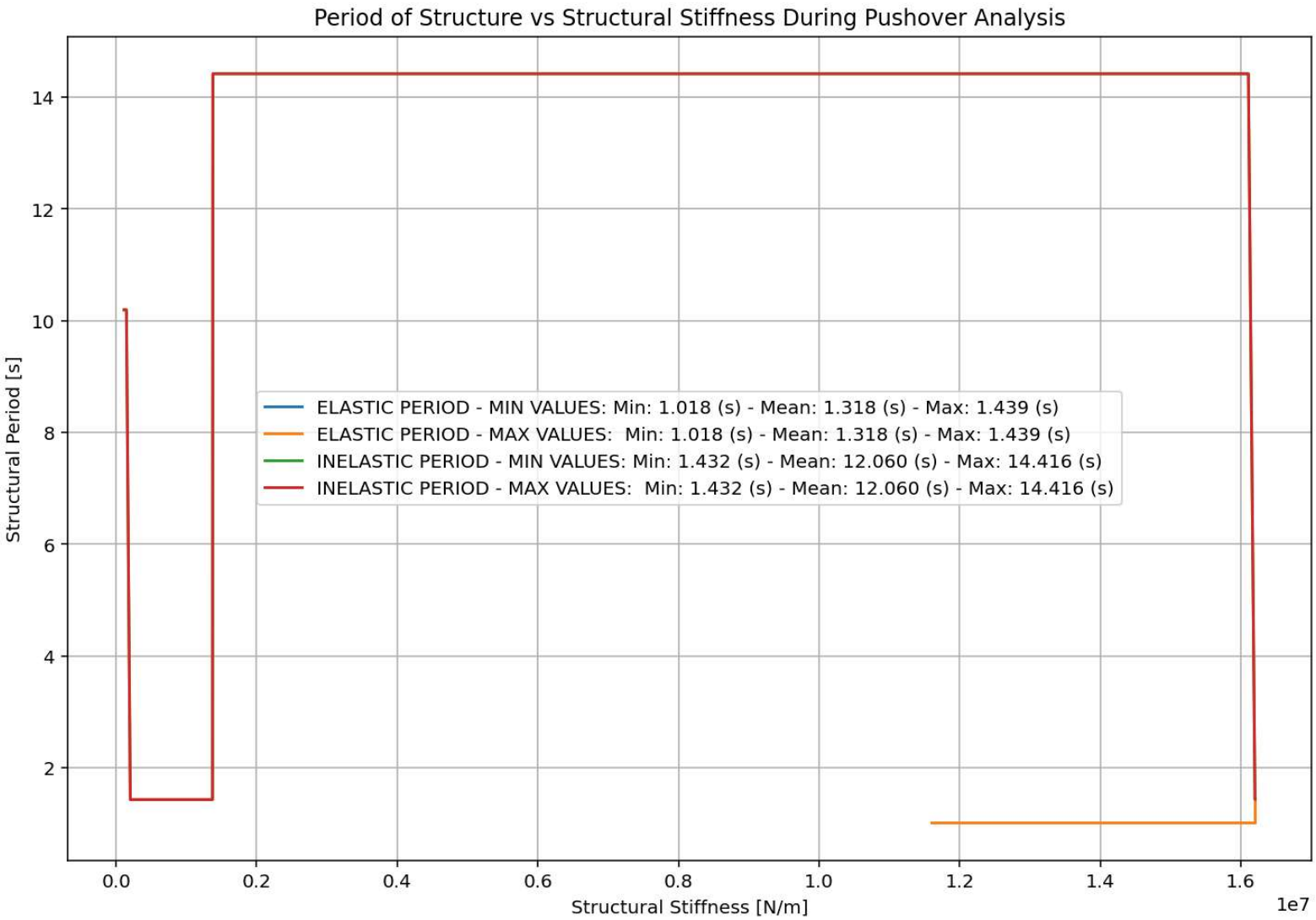
IPython Console | Variable Explorer | Plots | Help | Debugger | History | Files

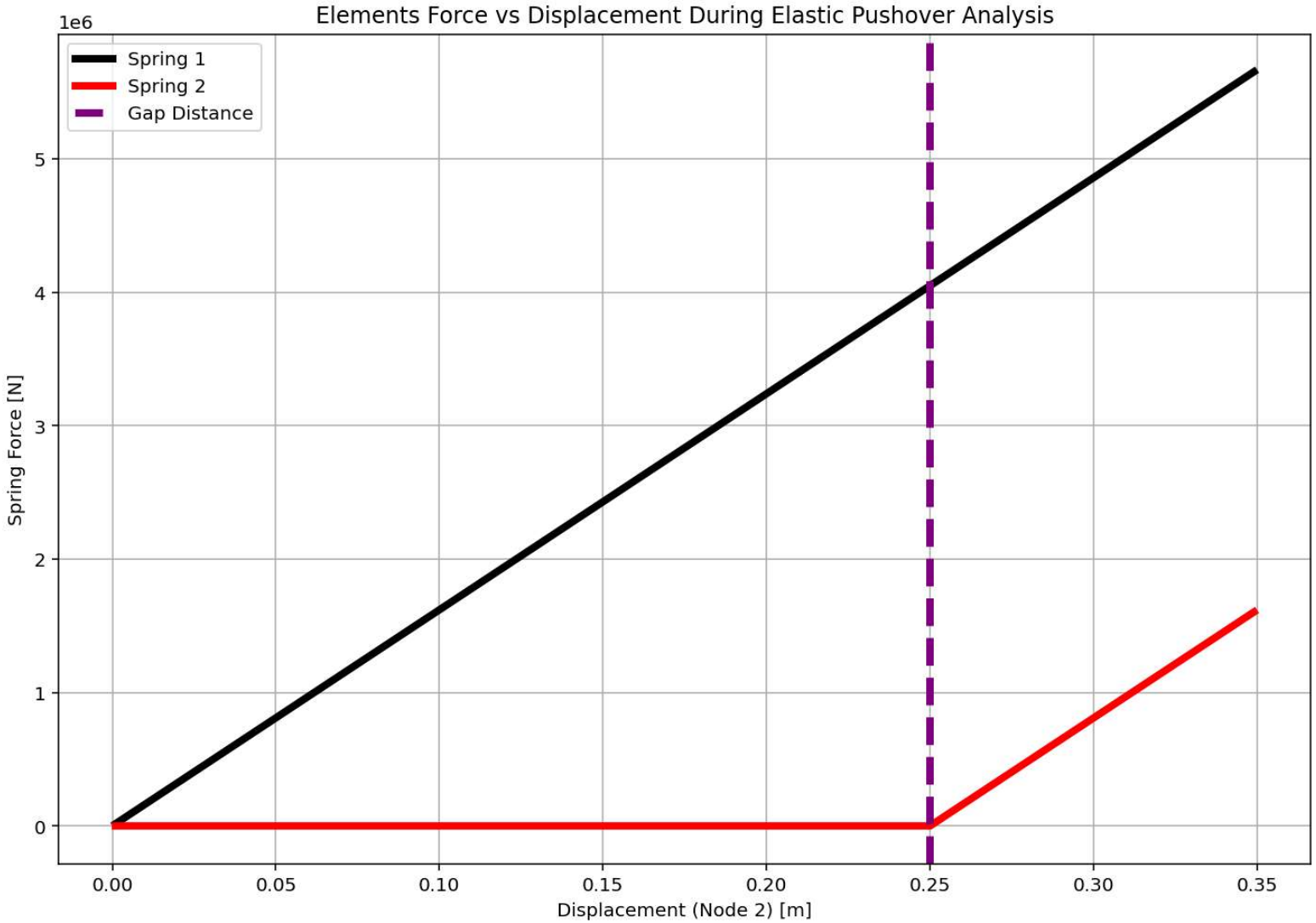
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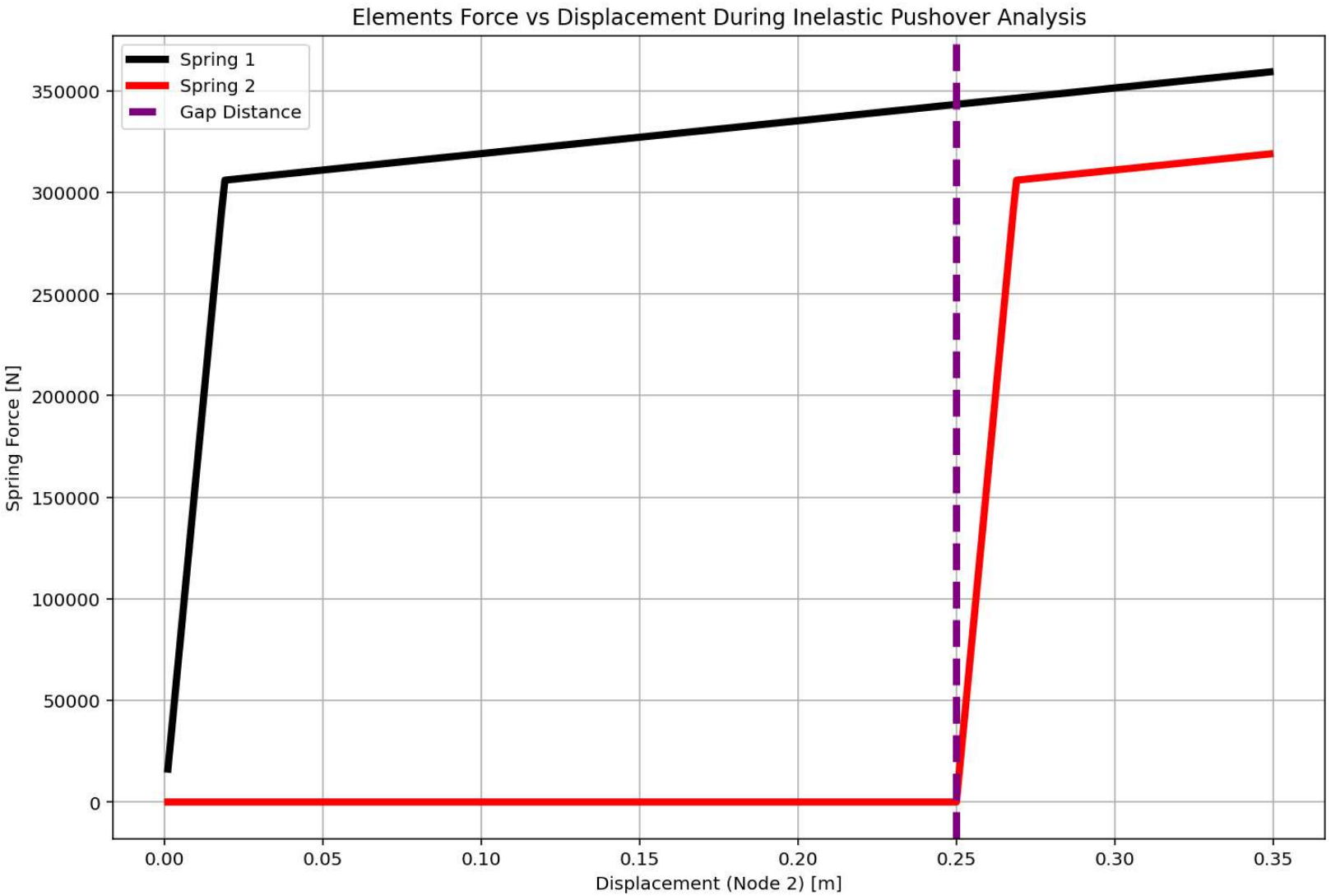

Force-Displacement Diagrams for 4 Different Inelastic Springs
Force [N]
Displacement [m]
150000
100000
50000
0
−50000
−100000
−150000
−0.4
−0.2
0.0
0.2
0.4
0.6
Element 1 (Original) - Tension
Element 1 (Original) - Compression
Element 2 (Stiffer) - Tension
Element 2 (Stiffer) - Compression
Element 3 (Softer) - Tension
Element 3 (Softer) - Compression
Element 4 (Modified) - Tension
Element 4 (Modified) - Compression

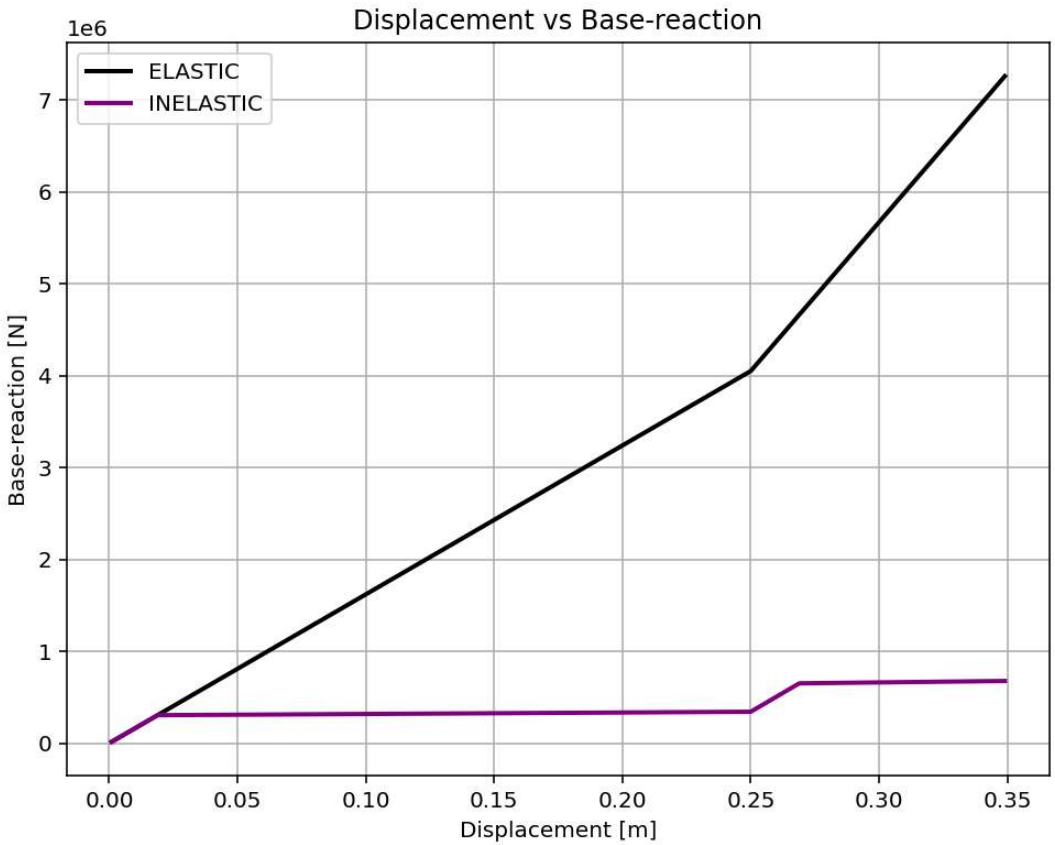

Element 1: Element 1 (Original)
Element 2: Element 2 (Stiffer)
Element 3: Element 3 (Softer)
Element 4: Element 4 (Modified)
Tension
Compression
Force [N]
Displacement [m]

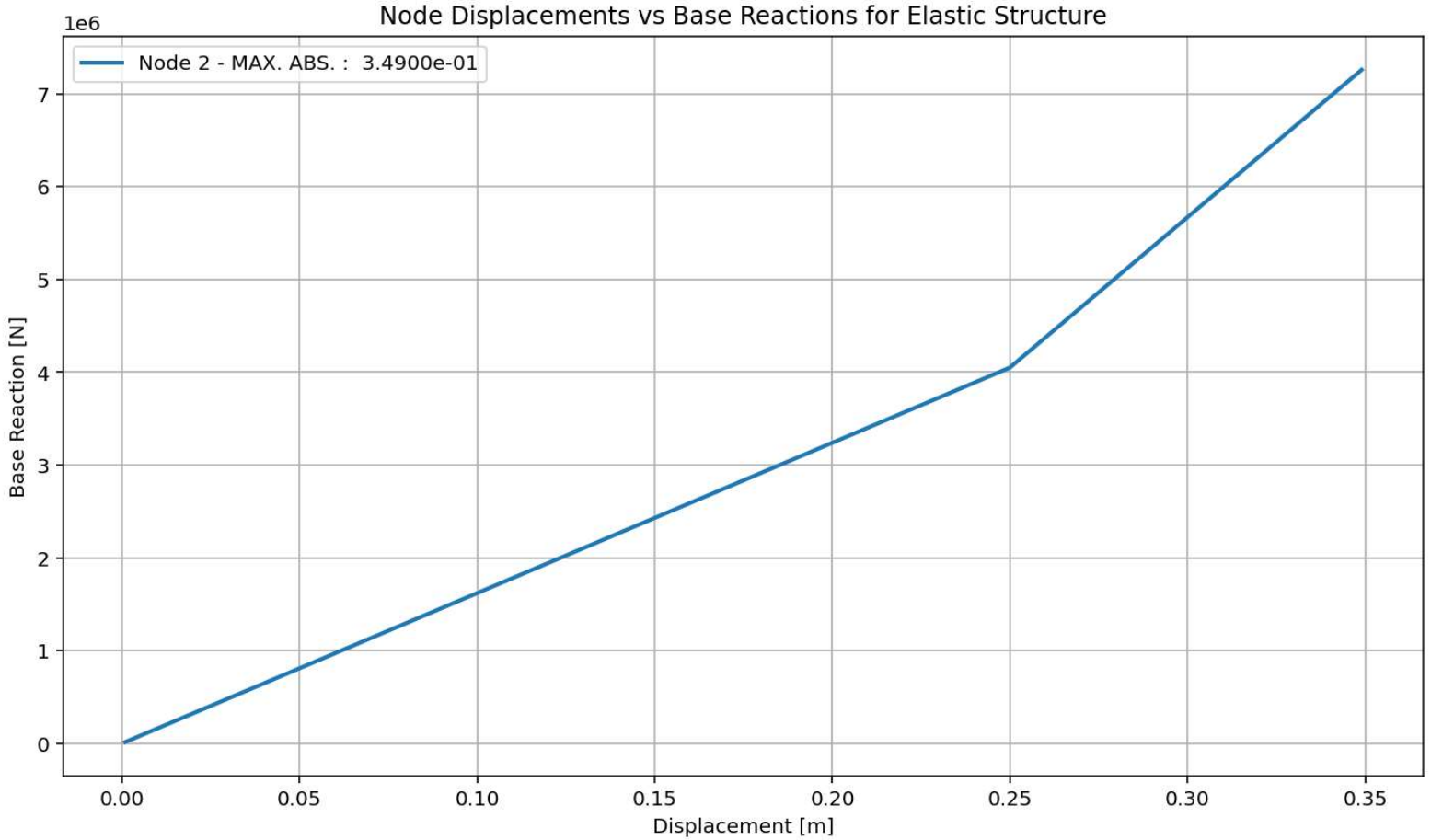

Period of Structure vs Damage level During Pushover Analysis
Structural Period [s]
Damage level [%]
ELASTIC PERIOD - MIN VALUES: Min: 1.018 (s) - Mean: 1.318 (s) - Max: 1.439 (s)
ELASTIC PERIOD - MAX VALUES: Min: 1.018 (s) - Mean: 1.318 (s) - Max: 1.439 (s)
INELASTIC PERIOD - MIN VALUES: Min: 1.432 (s) - Mean: 12.060 (s) - Max: 14.416 (s)
INELASTIC PERIOD - MAX VALUES: Min: 1.432 (s) - Mean: 12.060 (s) - Max: 14.416 (s)

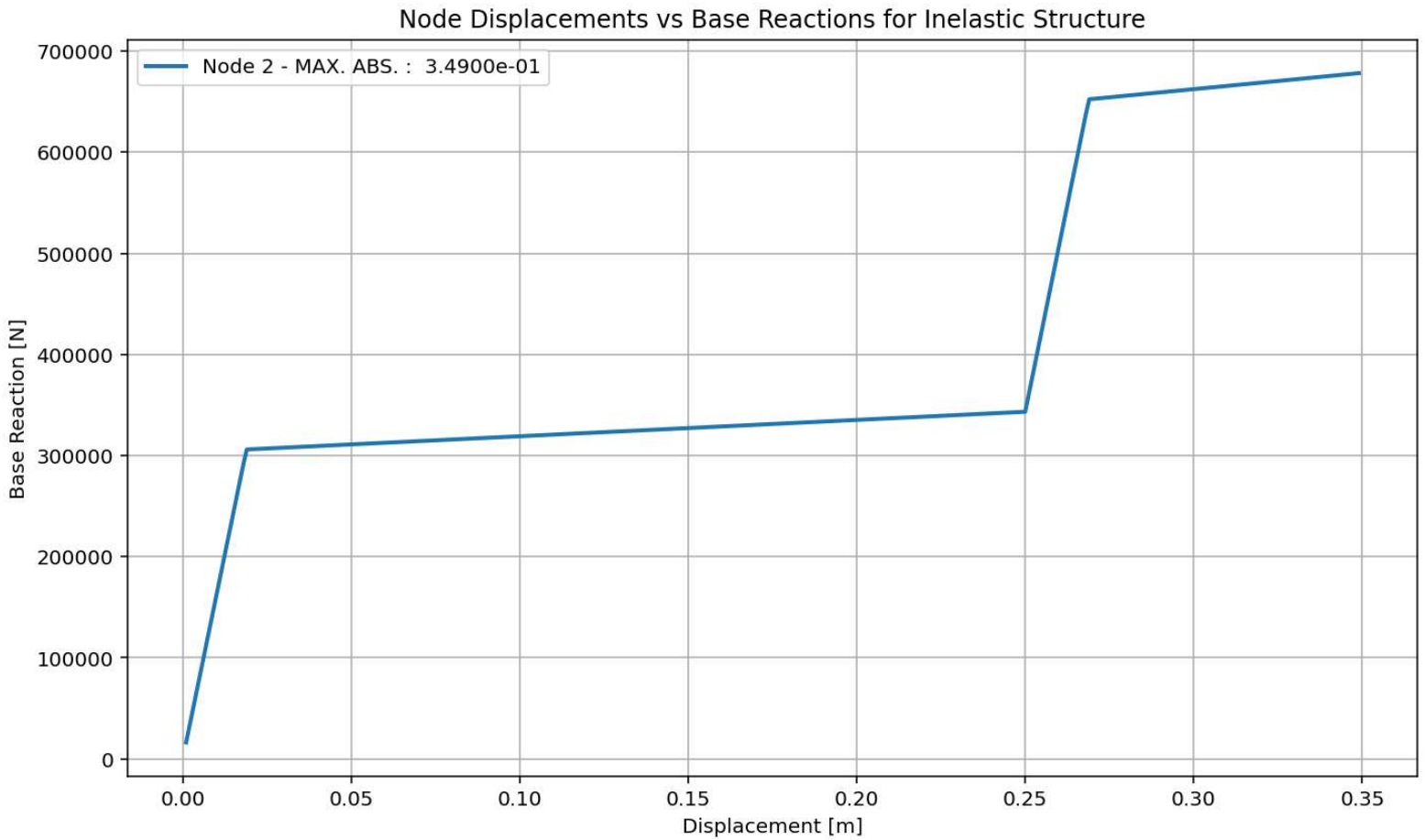

Period of Structure vs Structural Stiffness During Pushover Analysis
Structural Period [s]
Structural Stiffness [N/m]
1e7
ELASTIC PERIOD - MIN VALUES: Min: 1.018 (s) - Mean: 1.318 (s) - Max: 1.439 (s)
ELASTIC PERIOD - MAX VALUES: Min: 1.018 (s) - Mean: 1.318 (s) - Max: 1.439 (s)
INELASTIC PERIOD - MIN VALUES: Min: 1.432 (s) - Mean: 12.060 (s) - Max: 14.416 (s)
INELASTIC PERIOD - MAX VALUES: Min: 1.432 (s) - Mean: 12.060 (s) - Max: 14.416 (s)


Elements Force vs Displacement During Elastic Pushover Analysis
1e6
Spring 1
Spring 2
Gap Distance
Spring Force [N]
0
1
2
3
4
5
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
Displacement (Node 2) [m]


Elements Force vs Displacement During Inelastic Pushover Analysis
Spring 1
Spring 2
Gap Distance
Spring Force [N]
350000
300000
250000
200000
150000
100000
50000
0
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
Displacement (Node 2) [m]


Displacement vs Base-reaction
1e6
ELASTIC
INELASTIC
Base-reaction [N]
Displacement [m]
0
1
2
3
4
5
6
7
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35


Node Displacements vs Base Reactions for Elastic Structure
1e6
Node 2 - MAX. ABS. : 3.4900e-01
Base Reaction [N]
Displacement [m]
0
1
2
3
4
5
6
7
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35


Node Displacements vs Base Reactions for Inelastic Structure
Node 2 - MAX. ABS. : 3.4900e-01
Base Reaction [N]
Displacement [m]
700000
600000
500000
400000
300000
200000
100000
0
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35